
Unsupervised Learning Project: VAE for Hybrid Music Clustering

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Abstract

1 This project implements an unsupervised learning pipeline where we explore
2 the effectiveness of Variational Autoencoders (VAE) for music clustering using
3 multimodal features: audio, lyrics and genre. The methods are implemented on
4 the GTZAN dataset comprising approximately 1,000 audio tracks categorized by
5 10 genres. In an attempt to tackle the limited dataset size, audio augmentations
6 such as time-stretching, pitch-shifting and noise injection are applied, expanding
7 the database to 3,996 samples and introducing data diversity to improve model
8 generalization. Each sample is assigned spectral audio features, genre-based lyric
9 embeddings and genre. A Conditional Variational Autoencoder (CVAE) is trained
10 on the modified data and made to produce latent features corresponding to a track's
11 audio features, lyrics and genre, and the latent features are clustered using K-Means.
12 This CVAE-based clustering is quantitatively compared against PCA-based cluster-
13 ing, Autoencoder-based clustering, and direct spectral feature clustering in terms
14 of 4 clustering metrics Silhouette Score, Adjusted Rand Index (ARI), Normalized
15 Mutual Information (NMI), and cluster purity. The CVAE-based clustering scores
16 the highest Silhouette Score (0.0206) and the highest ARI (3.2586 $\times$ 10), indicat-
17 ing well-separated clusters and higher similarity between predicted clusters and
18 ground truth labels. However, it produces the lowest NMI (0.0046) and cluster
19 purity (0.1226), indicating weaker alignment between clusters and genre labels.
20 Comparatively, PCA-based clustering attains the highest NMI (0.0383) and cluster
21 purity (0.1857), indicating stronger alignment between clusters and genres.

22 1 Introduction

23 With digital music platforms growing popular among all ages of people day by day, the task of
24 analyzing and sorting music now demands more attention than ever. Music clustering contributes to
25 this by grouping songs sharing common characteristics such as genre, lyrics, melody, rhythm, beat and
26 so on, useful in applications such as music recommendation and playlist generation. Unsupervised
27 clustering, such as in this project, is more independent and flexible in discovering latent musical
28 representations due to the absence of labels, which cannot be said for supervised clustering.

29 Traditional music clustering methods are mainly conducted on extracted audio features, such as
30 Mel-Frequency Cepstral Coefficients (MFCCs), which they, more often than not, struggle to capture
31 the complex, nonlinear patterns of. In addition to that, music consists of multimodal features that
32 cannot be fully captured by acoustic features alone.

33 In recent works, Variational Autoencoders (VAEs), have proven to be quite effective in learning
34 compact and meaningful latent representations from high-dimensional features. Their particular
35 framework produces smooth and structured latent spaces, especially suitable for clustering. Con-
36 ditional VAEs (CVAEs) further develops on this capability by incorporating information such as

genre, which computes cleaner and disentangled latent representations, hence, improving clustering quality and interpretability. They provide a structured way to fuse heterogeneous modalities into disentangled latent space which remains a challenge for the traditional approaches.

This study explores hybrid music clustering by implementing a Conditional Variational Autoencoder-based clustering approach on audio and lyrical features, conditioned with genre labels. This approach is compared with several baseline methods, such as PCA, Autoencoders, and direct feature clustering in terms of clustering metrics such as Silhouette Score, ARI, NMI and cluster purity, providing a comprehensive evaluation of label-based cluster quality and alignment with ground truth labels

2 Related Work

Variational Autoencoders (VAEs) are popular towards the unsupervised learning of small latent representations of high-dimensional data. It has been demonstrated in previous papers that VAEs can be effective in representation learning in tasks of audio and music analysis because they can represent data distributions and enforce regularization of latent space. Convolutional VAEs and Conditional VAEs are extensions that enhance the performance of structured inputs (e.g. spectrograms) in capturing local temporal and spectral features and use side information (e.g. class labels) respectively to control the organization of latent space.

Clustering is a popular method of studying learnt representations. Simple and scalable methods like K-Means are commonly employed and hierarchical methods like Agglomerative Clustering can allow the multi-level structure to be discovered. Density-based algorithms like DBSCAN can be helpful in detecting arbitrary-shaped clusters and noise without a given number of clusters. These methods are commonly used on learned embeddings, as opposed to raw features, in an effort to enhance cluster separability.

The most common audio features that music representation learning depends on include MFCCs and spectrograms, and textual features based on the lyrics of a song. Recent researches have researched multimodal approach in the combination of audio and lyrics to obtain the acoustic and semantic information. This hybrid representation was demonstrated to be superior to unimodal features in music similarity and clustering tasks, which is why it is used in this study as joint audio-text embeddings.

3 Method

3.1 Variational Autoencoders

3.1.1 Basic Variational Autoencoder

A basic Variational Autoencoder (VAE) is a probabilistic generative model which takes in a high-dimensional input and outputs a low-dimensional latent representation of that input. It consists of an encoder and a decoder. The encoder maps the input to the parameters of a probability distribution in latent space, typically a multivariate Gaussian characterized by a mean vector and a log-variance vector. VAE learns a distribution over latent space, modeling uncertainty. Using the reparameterization trick, a latent vector is sampled from this distribution which is then fed into the decoder to reconstruct the input. Training a VAE involves optimizing a loss function which includes a reconstruction term and a Kullback–Leibler divergence term. This outputs smooth latent vectors suited for clustering and representation learning.

3.1.2 Convolutional Variational Autoencoder

A Convolutional Variational Autoencoder builds upon the basic VAE by incorporating convolutional layers in the encoder and decoder, suited for structured data such as images, spectrograms, or time–frequency representations. Using convolutional filters, the encoder finds patterns while progressively reducing dimensionality through strided convolutions. These feature maps are then flattened and mapped to latent vectors. The decoder uses transposed convolutional layers to reconstruct the input. Convolutional VAEs are better in terms of parameter efficiency and preservation of

85 spatial relationships than fully connected VAEs. This makes them more capable to learn hierarchical
86 representations that capture both low-level and high-level structural information in the data.

87 **3.1.3 Conditional Variational Autoencoder**

88 A Conditional Variational Autoencoder (CVAE) incorporates auxiliary information into the process
89 due to which model can learn condition-aware latent representations. Here, both the encoder
90 and the decoder are conditioned on an additional variable, such as a class label or attribute vector.
91 Consequently, the latent space is structured both by the input data and the conditioning information. A
92 conditional distribution of the latent variables is learned to produce disentangled latent representations.
93 This design is particularly useful when the variations in data heavily depend on known attributes,
94 encouraging the latent space to reflect these variations explicitly.

95 **3.2 Feature extraction**

96 Feature extraction serves as the bridge between raw multimedia data and machine learning models.
97 For audio data, Mel-Frequency Cepstral Coefficients (MFCCs) are widely used as audio features
98 that represent the short-term power spectrum of sound while incorporating a perceptual mel-scale
99 that reflects human auditory sensitivity. They are produced by dividing the audio into short frames
100 and mapping frequencies to the mel scale on which a discrete cosine transform is eventually applied,
101 producing a compact representation that captures timbral and spectral characteristics of the audio
102 over time.

103 Lyrics provide semantic and contextual information. As neural networks cannot directly process raw
104 text, lyrics are converted into numerical vectors using text embedding techniques. Sentence-level
105 embeddings encode semantic information into corresponding vectors, allowing information such
106 as themes and emotions to be captured in a continuous space. These embeddings depict numerical
107 comparisons among the music samples in terms of their lyrical content.

108 **3.3 Clustering Methods**

109 **3.3.1 K-Means**

110 K-Means clustering is a partition-based unsupervised learning algorithm. It divides a dataset into a
111 fixed number of clusters. The algorithm initializes a set of cluster centroids and iteratively assigns
112 each data point to the cluster that the nearest centroid belongs to. A distance metric, typically
113 Euclidean distance, is used to calculate the distance between each data point and a centroid. Centroids
114 are then updated as the mean of all points within each cluster. This process continues until cluster
115 assignments no longer change significantly. K-Means is computationally efficient and effective for
116 producing well-separated clusters, but it requires the number of clusters to be specified in advance
117 and is sensitive to outliers.

118 **3.3.2 Agglomerative Clustering**

119 Agglomerative Clustering is a hierarchical clustering technique that builds a tree-like structure of
120 clusters. At first, each data point is considered to be an individual cluster. At every step, the two
121 closest clusters are merged based on a chosen linkage criterion, such as single or average linkage.
122 This merging continues until a good clustering level is reached. Agglomerative Clustering is flexible
123 and does not require the number of clusters to be specified beforehand, but can be computationally
124 expensive for large datasets.

125 **3.3.3 DBSCAN**

126 DBSCAN is a clustering algorithm based on density that produces dense clusters of data separated by
127 sparse regions. It groups points that have enough neighbors within a specified radius, and classifies
128 points that do not meet this density requirement as noise. DBSCAN is particularly effective at
129 discovering clusters of arbitrary shape and handling outliers. It does not require a predefined number
130 of clusters, however, its performance depends heavily on chosen density parameters.

131 4 Experiments

132 4.1 Dataset

133 The experiments are conducted using the widely adopted GTZAN music dataset, a standard bench-
134 mark for music analysis tasks. The original dataset comprises of approximately 1,000 audio tracks
135 of 30 seconds long each, uniformly categorized by 10 music genres. From the original audio files,
136 MFCC features and lyric embeddings are extracted to create a processed dataset suitable for varia-
137 tional representation learning. The resulting feature dataset is publicly available at [https://drive.](https://drive.google.com/drive/folders/1J3F5L6XsmgVoyoJ6JdhMZfmwGDJ70QIM?usp=sharing)
138 [google.com/drive/folders/1J3F5L6XsmgVoyoJ6JdhMZfmwGDJ70QIM?usp=sharing](https://drive.google.com/drive/folders/1J3F5L6XsmgVoyoJ6JdhMZfmwGDJ70QIM?usp=sharing).

139 4.2 Pre-processing

140 In order to increase data diversity and improve model generalization, the audio samples are augmented
141 using techniques such as time-stretching, pitching and adding noise, resulting in a total of 3,996
142 samples.

143 Each audio track is first transformed into Mel-Frequency Cepstral Coefficients (MFCCs) to capture
144 time-frequency characteristics relevant to human auditory perception. Specifically, 20 MFCC
145 coefficients are extracted per frame of a track. They are then truncated to a fixed number of frames
146 (1,172), producing tensors of shape (3,996, 1,172, 20). This fixed-length representation ensures
147 compatibility across all learning architectures. These tensors are flattened into dimension 23,440
148 ($=1,172 \times 20$) which are used in the basic fully connected VAE and in the conditional VAE. For
149 convolutional architectures, the MFCCs are preserved in their two-dimensional form and reshaped to
150 (B, 1, 20, W), where B denotes the batch size, 1 is the single-channel input, 20 represents the number
151 of MFCC coefficients, and $W = 1,172$ corresponds to the temporal axis.

152 Lyrics are incorporated as a second modality by constructing genre-based lyrics for each track. For
153 every audio sample, lyrics associated with its genre are used to form a textual description. These
154 lyrics are transformed into embeddings of size 384, resulting in lyric feature vectors of shape (3,996,
155 384). These embeddings are concatenated with audio-derived features in hybrid learning setups.

156 Genre labels are encoded as one-hot vectors of length 10, corresponding to the 10 genres in the dataset.
157 These vectors are used as conditioning variables in the conditional VAE and are also concatenated
158 with audio and lyric feature vectors. All features are stored as 32-bit floating-point values.

159 4.3 Hyperparameter

160 All variational models employ a latent dimensionality of 32, allowing a consistent comparison across
161 architectures and clustering approaches. Training is performed using the Adam optimizer with a
162 learning rate of 1×10^{-3} . The basic VAE uses a batch size of 64, while the conditional VAE uses a
163 batch size of 32 due to the larger input dimensionality. The convolutional VAE also uses a batch size
164 of 32. Each model is trained for 30 epochs.

165 For the basic VAE, KL divergence warm-up is applied over the first 10 epochs, gradually increasing
166 the contribution of the KL term to stabilize early training. Clustering is performed as follows:
167 K-Means with 10 clusters (matching the number of genres) and multiple random initializations
168 ($n_{init} = 10-20$) to reduce sensitivity to initialization, DBSCAN with $\epsilon = 1.5$ and a minimum of 10
169 samples, and agglomerative clustering configured to produce 10 clusters. Dimensionality reduction
170 baselines, including PCA and autoencoders, reduce features to 32 dimensions to align with the VAE
171 latent space.

172 4.4 Training

173 For the easy task, a basic fully connected Variational Autoencoder is trained using audio-only features.
174 The flattened MFCC vectors of dimension 23,440 are passed through an encoder consisting of two
175 fully connected layers with 1,024 and 512 hidden units, respectively. The encoder outputs the mean
176 and log-variance vectors, each of size 32, representing the parameters of the latent Gaussian distribu-
177 tion. The decoder mirrors this structure with layers of sizes 512 to 1,024 to 23,440, reconstructing
178 the original MFCC features. Training optimizes a loss function combining mean squared error (MSE)
179 reconstruction loss and Kullback-Leibler divergence, with KL warm-up applied over the first 10

180 epochs. After training, the latent representations are obtained by extracting the mean vectors for all
181 samples.

182 For the medium task, a Convolutional Variational Autoencoder is trained on audio and lyrical
183 features jointly. Audio inputs are provided in the form $(B, 1, 20, W)$, where B is the batch size,
184 1 denotes a single input channel, 20 corresponds to the number of MFCC coefficients, and $W =$
185 1,172 represents the temporal dimension. The encoder consists of three convolutional layers with 16,
186 32, and 64 filters, respectively, each using a kernel size of 3×3 , stride 2, and batch normalization
187 followed by ReLU activation. The resulting feature maps are flattened and concatenated with 384-
188 dimensional lyric embeddings before being mapped to a 32-dimensional latent space. The decoder
189 uses a fully connected layer followed by three transposed convolutional layers to reconstruct the
190 MFCC representation, ensuring that the output matches the original input shape.

191 For the hard task, a Conditional Variational Autoencoder is trained on hybrid audio-lyric features
192 conditioned on genre labels. The input consists of flattened audio-lyric feature vectors of dimension
193 23,824, concatenated with a 10-dimensional one-hot genre vector. The encoder comprises two fully
194 connected layers with 512 and 256 units, followed by latent mean and log-variance layers of size
195 32. During decoding, the latent vector is concatenated with the same genre condition vector and
196 passed through symmetric decoder layers of sizes 256 to 512 to 23,824. This conditioning enforces
197 genre-aware latent representations. Training uses an MSE reconstruction loss and KL divergence,
198 and latent vectors are sampled using the reparameterization trick before being used for clustering and
199 visualization.

200 5 Results

201 5.1 Clustering metrics

202 Different representation learning and clustering methods were compared in terms of standard cluster-
203 ing statistics, such as, Silhouette Score, Davies Bouldin Index (DBI) and Adjusted Rand Index (ARI),
204 Normalized Mutual Information (NMI) and cluster purity, depending on whether or not ground-truth
205 genre labels were available.

206 PCA-K-Means baseline clustering resulted in Silhouette Score of 0.0036, NMI, and ARI values of
207 0.0383 and 0.0166 respectively. This shows that there is a lack of distance between clusters, and a low
208 level of congruence with the actual genre names. The simplest VAE that was trained on audio-only
209 features of MFCC produced extremely poor performance in clustering, with a Silhouette Score of
210 -1.0 and an ARI of -1.0 , showing that the latent space is not able to form high-quality clusters.

211 Convolutional VAE, which also simultaneously predicted audio MFCCs and lyric embeddings,
212 scored significantly higher on clustering than did the baselines. It was used in combination with
213 K-Means clustering with a Silhouette Score of 0.1951 and Davies Bouldin Index of 1.8182. When
214 Agglomerative clustering was used on the same latent representations, it gave a comparable Silhouette
215 Score value of 0.1923 and a DaviesBouldin Index of 1.8357. DBSCAN failed to give valid clustering
216 results on this latent space.

217 Conditional VAE (CVAE) reached the Silhouette Score of 0.0206, NMI of 0.0046 and ARI of
218 3.26×10^{-5} . The purity of corresponding cluster was 0.1226.

219 Further comparisons on the baseline were performed with the help of representations in autoencoders
220 and pure clustering on the audio, lyrics, and genre concatenated features. Silhouette Score is -0.0030 ,
221 and it is -0.0028 with the auto encoder and direct feature clustering respectively. The two approaches
222 gave lower NMI and ARI and purity scores than the Convolutional VAE and CVAE algorithms.

223 5.2 Latent space visualizations

224 Latent space visualization was created with the t-SNE to map the latent vectors of high dimensions to
225 two dimensions. In the case of the Conditional VAE, it was done by creating visualizations of true
226 genre labels and K-Means cluster assignments. These visualizations demonstrate apparent patterns of
227 grouping in the latent space, and areas of the space have more dense samples. The t-SNE plots when
228 colored by the clusters predicted by them will show distinguishable regions that can be attributed to
229 certain cluster assignments. In the case of being colored in actual genre terms, some overlap between
230 genres can be traced.

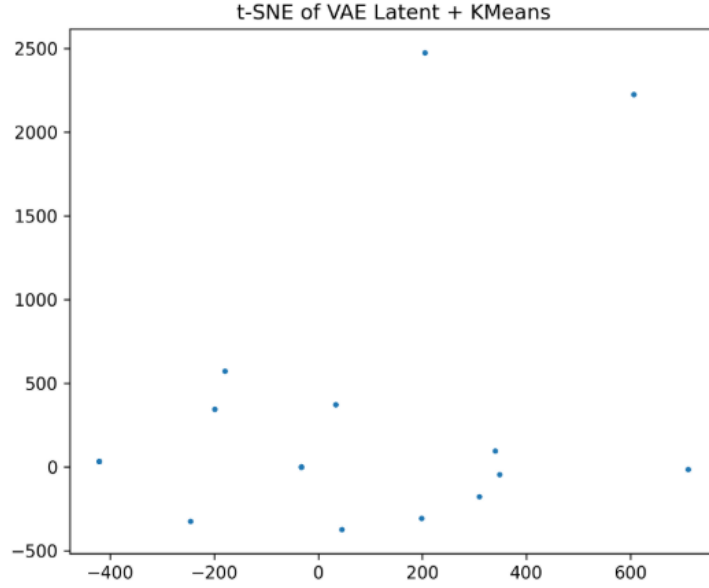


Figure 1: t-SNE visualization of VAE-reduced features from audio only

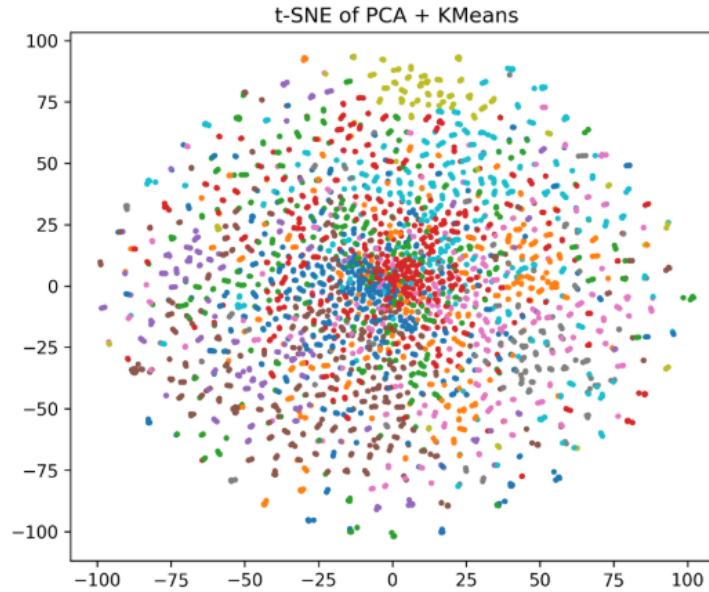


Figure 2: t-SNE visualization of PCA-reduced features from audio only

231 5.3 Comparison with baseline methods

232 Comparison of all of the methods reveals that learned latent representations tend to perform better
 233 than direct clustering on raw or linearly reduced representations. PCA, autoencoder based repre-
 234 sentations and direct feature clustering had fairly low and similar clustering scores in all measures.
 235 Convolutional VAE obtained the highest silhouette scores and lowest Davies Bouldin indices, whereas
 236 the Conditional VAE produced quantifiable genre consistency in the form of NMI and purity scores.
 237 These findings can be used as quantitative foundations on further analysis and interpretation discussed
 238 in the following section.

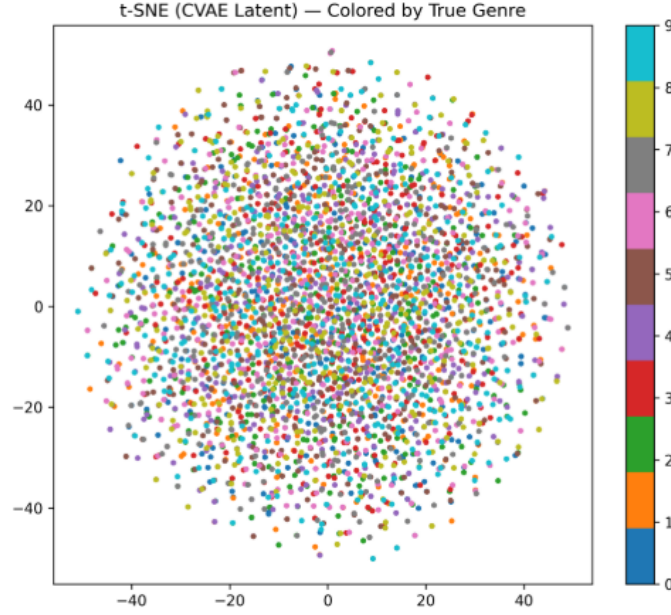


Figure 3: t-SNE visualization by genre of CVAE-reduced features from audio, lyrics, and genre

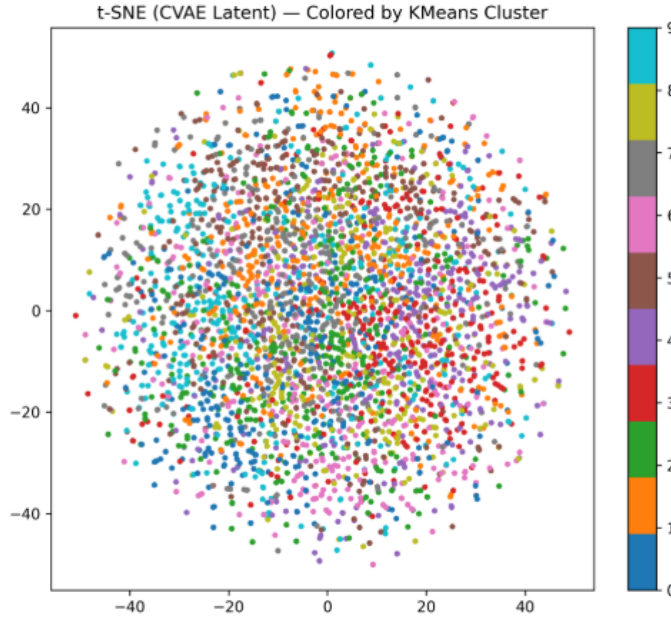


Figure 4: t-SNE visualization by clusters of CVAE-reduced features from audio, lyrics, and genre

6 Discussion

6.1 Analysis of Results

The experimental findings prove that the quality of clustering is highly dependent on the representation learning methodology. PCA, direct feature clustering, and conventional autoencoders always yielded low silhouette, NMI, and ARI scores, suggesting that they had poor cluster separation and poor alignment with genre labels. This implies that linear dimensionality reduction and shallow reconstruction-based models cannot be used to represent the complex and non-linear relationships existing in multimodal music data.

247 The simplistic VAE that was being trained on audio features only showed a very poor clustering
248 performance. This is shown by the negative silhouette score and ARI which shows that the latent
249 space did not retain meaningful structure that is important to genre differentiation. The implication
250 of this result is that high-dimensional audio features are hard to model without explicitly trying to
251 capture local time-frequency patterns.

252 On the contrary, the Convolutional VAE demonstrated the best clustering performance of all the
253 considered models. The scores in the silhouette are comparatively higher whilst the DaviesBouldin
254 indices are lower indicating enhanced compactness and separation of clusters in the latent space.
255 Convolutional layers were presumably necessary to provide the model with a more accurate portrayal
256 of local spectral structure in MFCC representations, with the addition of lyric embeddings providing
257 a additional semantic data. The fact that the K-Means and Agglomerative clustering gave similar
258 results on this latent space also suggests that the learned representations were structurally consistent.

259 The Conditional VAE showed small gains over the baseline approaches but it failed to outperform
260 the Convolutional VAE. Although genre label conditioning brings in guided training, the clusters
261 that still emerged were quite overlapping. This implies that genre information might not be adequate
262 enough to completely determine latent elements in music data particularly where the genres overlap
263 in terms of style or acoustics.

264 DBSCAN did not give valid clusters over the learned latent representations. This observation is
265 similar to the density-based methods that are sensitive to the choice of parameters and different
266 sample density, especially in high-dimensional latent space or partially overlapping latent space.

267 6.2 Interpretation of clusters

268 The observed clusters in the latent space reflect partial grouping based on shared acoustic and semantic
269 characteristics rather than strict genre boundaries. Visualizations indicate that some genres form
270 denser regions, while others overlap substantially. This aligns with the inherent subjectivity and
271 fluidity of music genres, where tracks often exhibit cross-genre traits. Clusters produced by the
272 Convolutional VAE appear more compact and spatially separated compared to those generated by
273 baseline methods. This suggests that combining convolutional audio modeling with lyric embeddings
274 enables the latent space to encode higher-level musical attributes. However, the lack of sharp genre
275 separation implies that the clusters may represent stylistic similarities, instrumentation patterns, or
276 lyrical themes rather than discrete genre classes.

277 6.3 Limitations

278 Several limitations affect the findings of this study. First, although data augmentation increased the
279 number of samples, the dataset remains limited in diversity and may not fully capture the variability
280 within each genre. Augmentation techniques may also introduce redundancy, which can influence
281 clustering behavior. Second, due to the unavailability of original lyrics, genre-based lyrics were
282 assigned to each track, hence, tracks belonging to the same genre produce similar lyric embeddings
283 which can weaken the contribution of textual features. Additionally, lyric embeddings may not
284 adequately capture musical context when used independently of audio features. Third, clustering
285 evaluation relies on genre labels as a proxy for ground truth, despite the fact that genres are inherently
286 ambiguous and overlapping. As a result, low ARI or NMI scores do not necessarily imply poor
287 representation quality but may instead reflect limitations of the labeling scheme. Finally, the models
288 were trained and evaluated using fixed architectural and hyperparameter choices. Further tuning,
289 larger datasets, or alternative conditioning variables could lead to improved performance and more
290 interpretable clusters.

291 7 Conclusion

292 This paper investigated unsupervised and semi-supervised music clustering on representation learning
293 with Variational Autoencoders on a variety of difficulty levels. Conventional baseline techniques like
294 PCA, direct feature clustering, and regular autoencoders were not very effective as they generated
295 weak cluster separation and low compatibility with genre labels. These findings suggest that the
296 straightforward linear or reconstruction-oriented models cannot be used to represent the complicated
297 nature of music data.

298 The simple fully connected VAE which was trained with audio only was unable to form meaningful
 299 clusters as indicated by negative or close to negative clustering scores. This underscores the difficulty
 300 of the task of representing high-dimensional audio representations by not learning local spectral
 301 patterns directly. Convolutional architectures were introduced and significant improvements were no-
 302 ticed. Convolutional VAE, which incorporated convolutional audio modeling with lyric embeddings,
 303 had the most consistent best clustering performance of all the tested models, and better organization
 304 of latent space and compactness of clusters. The Conditional VAE also realized small incremental
 305 improvements compared to the baseline methods with the use of genre information in training.
 306 Nevertheless, its performance in clustering was not very high, implying that genre conditioning does
 307 not ensure that the latent representations are well differentiated, especially because of the overlap and
 308 ambiguity in the music genres.

309 In general, the results indicate that convolutional designs and integration of multi-modes are more
 310 useful when clustering music without supervision compared to linear and full connected designs.
 311 Meanwhile, the findings prove that genre clustering in music is a difficult endeavor because of seman-
 312 tic ambiguity and limited data as well as subjective categorization. Further research may focus on
 313 more lyric representations, stronger audio capabilities, and larger and more varied datasets to enhance
 314 generalization. Exploration of other conditioning variables, including mood or instrumentation, and
 315 more sophisticated clustering algorithms and self-guided goals might also be beneficial and improve
 316 latent space organization and explainability. It is also a good direction of future research to extend
 317 the framework to work with hierarchical or multi-label genres.

318 **References**

- 319 [1] Kingma, D.P. & Welling, M. (2014) Auto-Encoding Variational Bayes. In *Proceedings of the 2nd Interna-*
 320 *tional Conference on Learning Representations (ICLR)*.
 321 [2] Doersch, C. (2016) Tutorial on Variational Autoencoders.
 322 [3] Kingma, D.P., Mohamed, S., Rezende, D.J. & Welling, M. (2014) Semi-supervised Learning with Deep
 323 Generative Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 27.
 324 [4] Bishop, C.M. (2006) *Pattern Recognition and Machine Learning*. Springer.
 325 [5] Tzanetakis, G. & Cook, P. (2002) Musical genre classification of audio signals. *IEEE Transactions on Speech*
 326 *and Audio Processing* **10**(5):293–302.